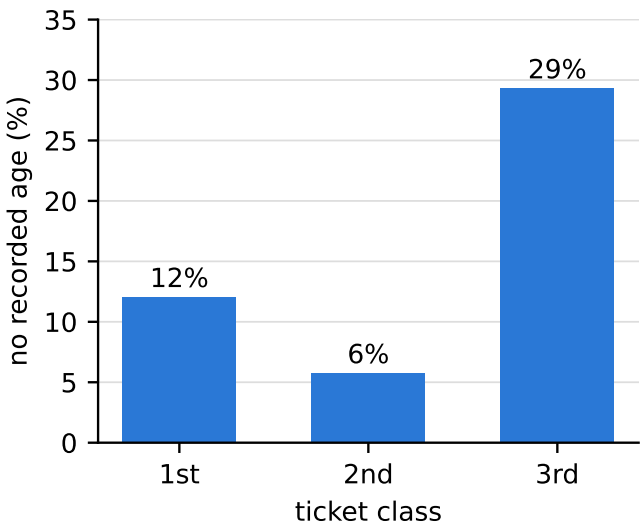
DATA 202 — Missing Data · Week 9 Wednesday

Same passengers as Monday, but all 1,309 of them: 263 have no recorded age. Your professor will tell you when to do each part.

1 · Who lost their age?



Share who survived:

class	age recorded	no age
1st	64%	49%
2nd	44%	25%
3rd	26%	24%

Talk with your neighbor: 3rd class lost the most ages, and in every class the passengers with no age survived less often. What story could explain **both** patterns?

2 · Name the mechanism

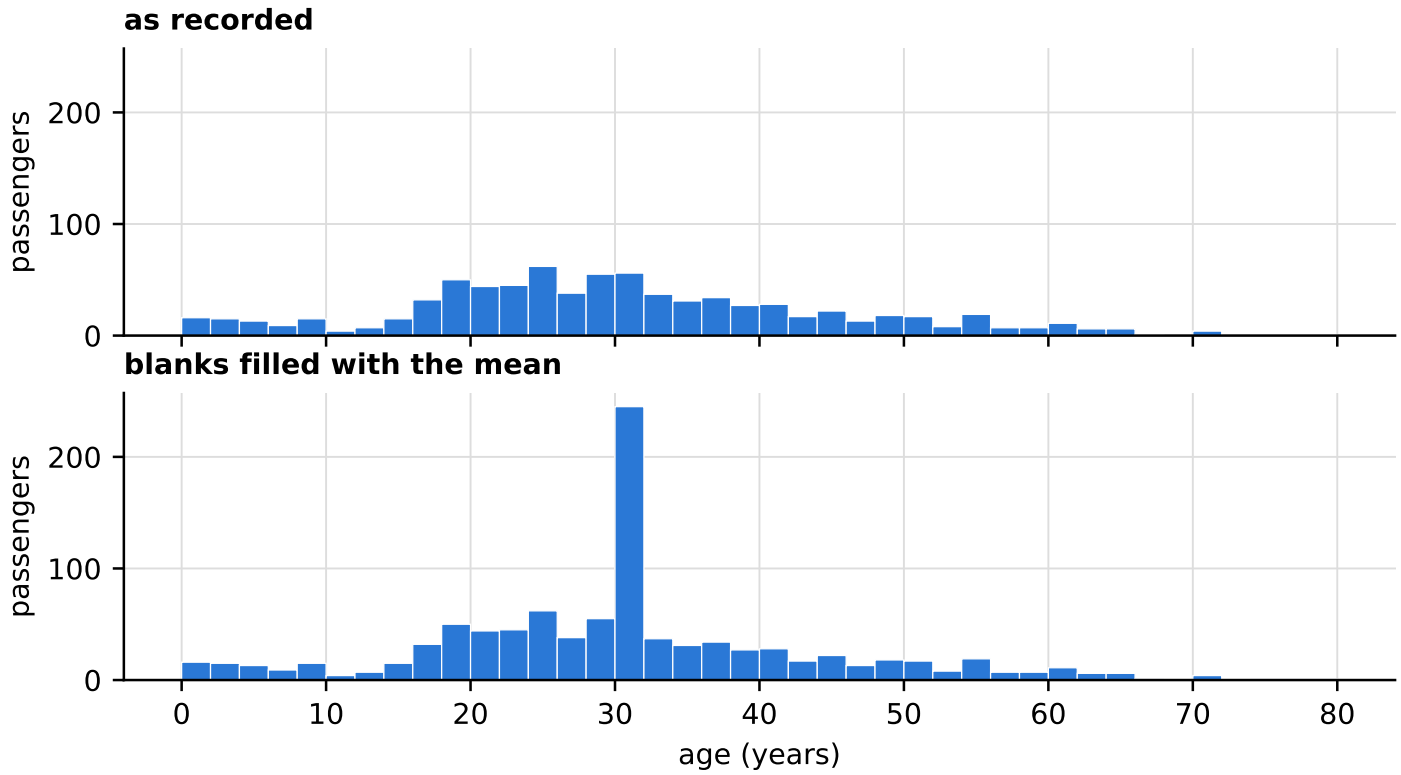
MCAR: pure chance. **MAR**: explained by columns we *can* see. **MNAR**: depends on the missing value itself, or on something never recorded.

Write MCAR, MAR or MNAR in each blank.

1. _____ A lab freezer breaks overnight and spoils a random rack of blood samples.
2. _____ To keep an online survey short, a randomly chosen half of the people never see the income question.
3. _____ Two *Titanic* passengers have no port: a clerk smudged their line in the register.
4. _____ A weather station's thermometer stops working below −30 °C.
5. _____ In a salary survey, the people with the highest salaries skip the salary question.
6. _____ A fitness watch records no steps on the days its owner leaves it at home: mostly lazy days.
7. _____ A clinic measures cholesterol only for patients over 40. Every patient's age is in the data.
8. _____ Men skip the "hours of sleep" question more often than women, however much they sleep. Sex is in the data.
9. _____ Cabin is blank for 98% of *Titanic* 3rd class: their tickets rarely listed one.
10. _____ Students who were failing a course dropped it, so their final grade is blank. Their midterm grades are in the data.
11. _____ The real missing *Titanic* ages from Part 1.

Talk with your neighbor: compare answers, then defend 10 and 11. For 10: what if the midterm column were missing? For 11: could the data ever settle it?

3 · The spike



The 981 training passengers' ages: as recorded (top), and after filling every blank with the mean age (bottom).

1. **Circle** what the imputer added.
2. **Talk with your neighbor:** median age is 39 in 1st class and 24 in 3rd. Whose blanks does one mean of 30 fit worse? What would you fill them with instead?

4 · Pick the fix

	fix	what it assumes about the blank rows
A	drop the rows with blanks	they are like everyone else
B	fill with one mean or median	they are like everyone else
C	fill from similar rows	other columns explain who is blank
D	C, plus a missing-indicator column	the blank itself says something

Write the best letter for each situation from Part 2.

1. _____ The freezer spoiled a random rack of blood samples (#1).
2. _____ Men skip the sleep question; sex is in the data (#8).
3. _____ Failing students dropped the course; their midterms are in the data (#10).
4. _____ The highest earners skip the salary question (#5).
5. _____ The real missing *Titanic* ages (#11).

Talk with your neighbor: for the salaries, does any letter bring the missing salaries back? What else could you do?